

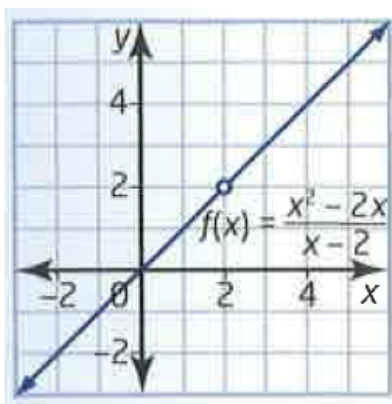
2.1/2.2 Multiplying and Dividing Rational Expressions – Lesson

MCR3U

What is a Rational Expression?

Rational Expression:

Example of a graph of a rational expression:



□ The open circle is used to represent a hole in the graph. This corresponds to any restrictions on the variable (denominator can't be 0).

Stating Restrictions

Note: rational expressions must be checked for restrictions by determining where the denominator is equal to _____. These restrictions must be stated when the expression is simplified.

□

Example 1: State the restrictions for the following rational expressions

a) $\frac{x+2}{x-2}$

b) $\frac{x+2}{(x-3)(x+4)}$

c) $\frac{5}{x(x+3)}$

Rule: We can cancel out ONLY when multiplying fractions

Rule: We can **NOT** cancel out when adding or subtracting fractions

Simplifying Rational Expressions

Example 2: Simplifying each expression and determine any restrictions on the variable.

a) $\frac{3x^2}{yx}$

b) $\frac{x-3}{x^2+3x-18}$

***Note:** factor where possible and then state restrictions before cancelling factors.*

c) $\frac{x^2+10x+21}{x+3}$

d) $\frac{x+1}{x^2+3x+2}$

$$\text{e)} \frac{x^2-9}{x^2+7x+12}$$

$$\text{f)} \frac{6x^2-7x-5}{3x^2+x-10}$$

Multiplying Rational Expressions

$$\text{a)} \frac{4x^2}{3x} \cdot \frac{12x^3}{2x}$$

1. factor where possible
2. cancel common factors
3. multiply numerators and denominators
4. state restrictions (throughout process)

$$\text{b)} \frac{4x+24}{x^2+8x} \cdot \frac{12x^2}{3x+18}$$

$$\text{c)} \frac{a^2+2a}{3a} \cdot \frac{20a^2}{5a^2+10a}$$

$$\text{d)} \frac{x+1}{2x} \cdot \frac{3x}{x^2+4x+3}$$

$$\text{e)} \frac{5x^2-13x+8}{x-7} \cdot \frac{1}{5x-8}$$

Dividing Rational Expressions

$$\text{a)} \frac{10ab^2}{4a} \div \frac{15a^2}{12b^2}$$

1. flip second fraction and change to multiplication
2. factor where possible
3. cancel common factors
4. multiply numerators and denominators
5. state restrictions (throughout process)

$$\text{b)} \frac{a^2+2a}{3a} \div \frac{5a^2+10a}{20a^2}$$

$$\text{c)} \frac{2x^2-8x}{x^2-3x-10} \div \frac{4x^2}{x^2-9x+20}$$